STORY WEBSITE DESIGN

- 5 mins at most, I will reminder when 30s left.

Jimmy
Sharon
Adrian
Emi
Cyrus
Wesley
Ben
William
Suman
Wisdom
Bryce

LORA

ISDN 3150 AI for Design, Yuan, 2026 Spring

PROBLEM: I WANT TO GENERATE X.

- Diffusion models are trained with vast public images, and we can generate subjects with words.
- But what if we want a specific subject with consistent design?
- But what if we want a specific style that is hard to describe?
- But what if we want to insert a new subject that was not in the dataset?

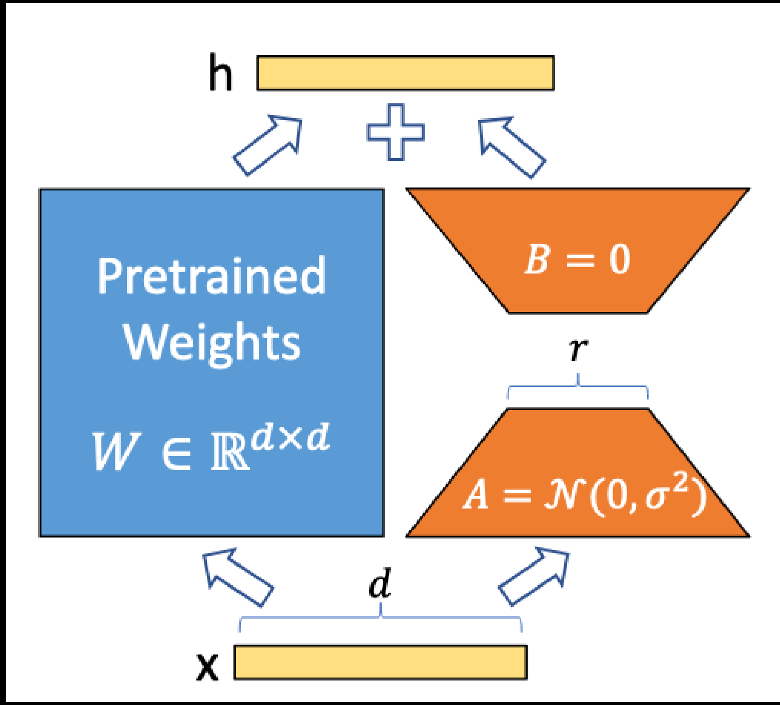
How about finetune the model?

Target Design

a cute stuffed boar doll with a black graudation gown and cap.

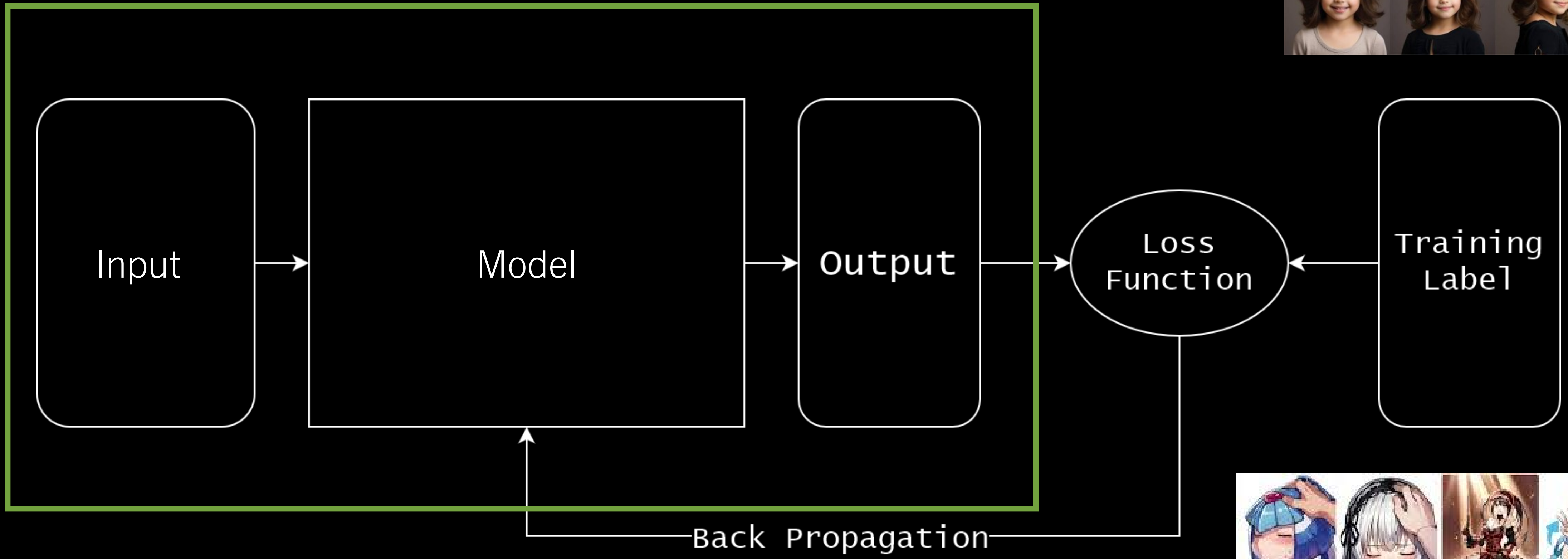
LORA (LOW-RANK ADAPTATIONS)

- Instead of directly training the model, train a smaller weight that “nudge” the model layer.
- Since we freeze the large model and only train the small weights, the training is more manageable.



DATA-DRIVEN MACHINE LEARNING

This is the inference stage



NEURAL NETWORKS

- 1st layer

$$\mathbf{h_1} = \mathrm{ReLU}(\mathbf{W_1 x} + \mathbf{b_1})$$

- 2nd layer

$$\mathbf{h_2} = \mathrm{ReLU}(\mathbf{W_2 h_1} + \mathbf{b_2})$$

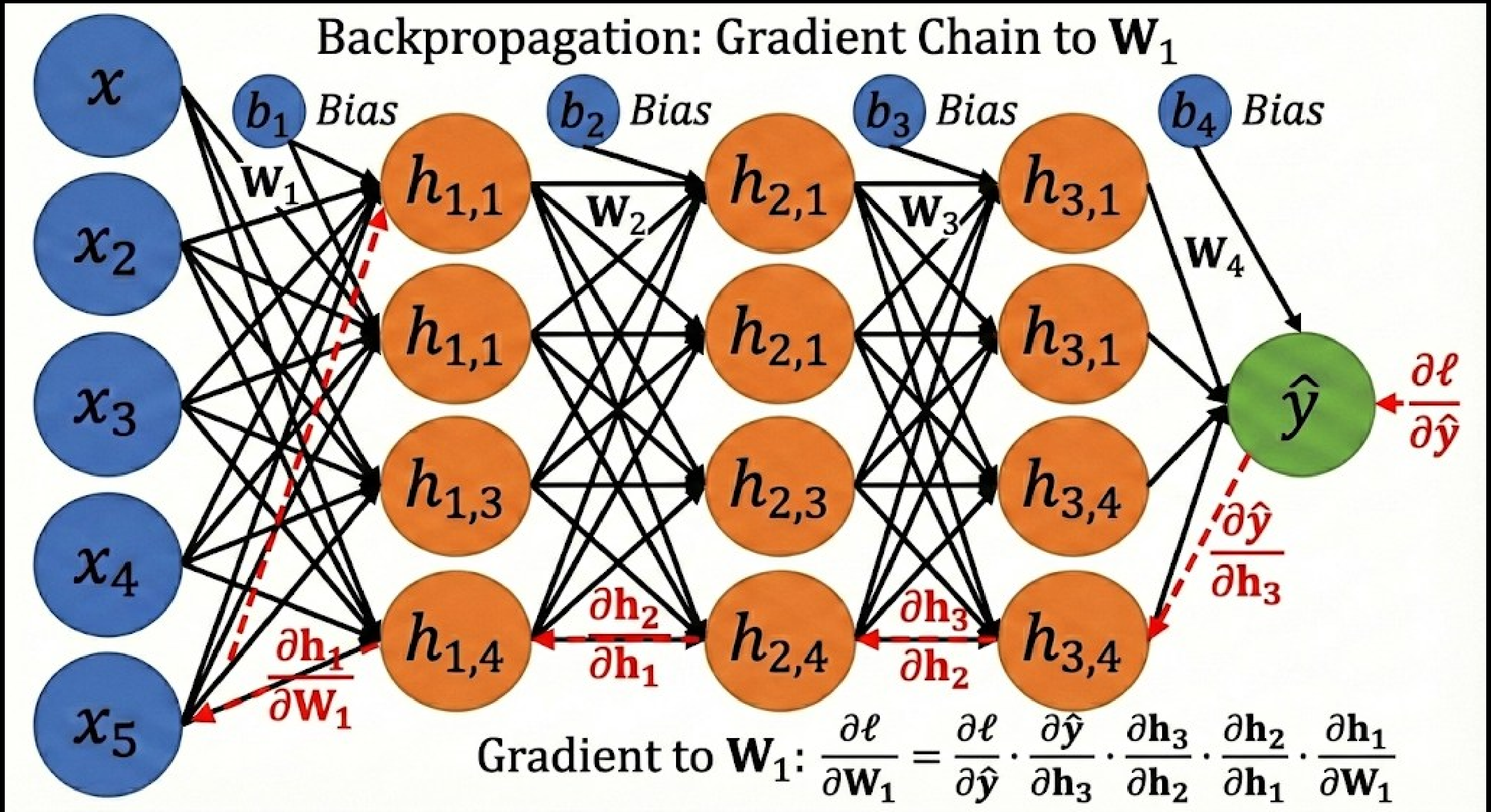
- 3rd layer

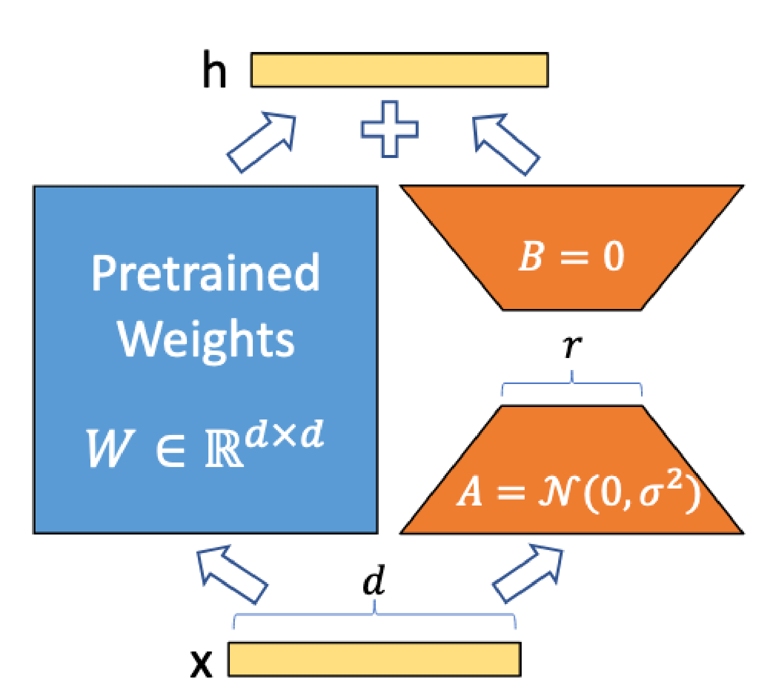
$$\mathbf{h_3} = \mathrm{ReLU}(\mathbf{W_3 h_2} + \mathbf{b_3})$$

- Output layer

$$\hat{y} = \sigma(\mathbf{W_4 h_3} + b_4)$$

NEURAL NETWORKS



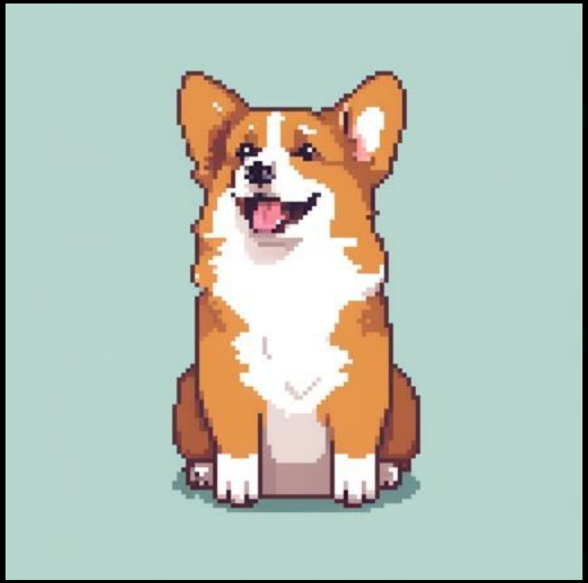


This idea also apply to arbitrary AI models!
As long as it has weight matrix for multiplication!

LORA

- LoRAs are very powerful, and they can range from subjects, styles, concepts and more!

Star Wars Characters



Pixel Art

Amateur Photography

https://civitai.com/models

LORA

- Same as dreambooth, since we are changing the model behavior, LoRAs often are trained with a trigger word.
- Without the trigger word present, the LoRA will not influence model a lot, preventing the effect appearing in unwanted places.

Trigger words:

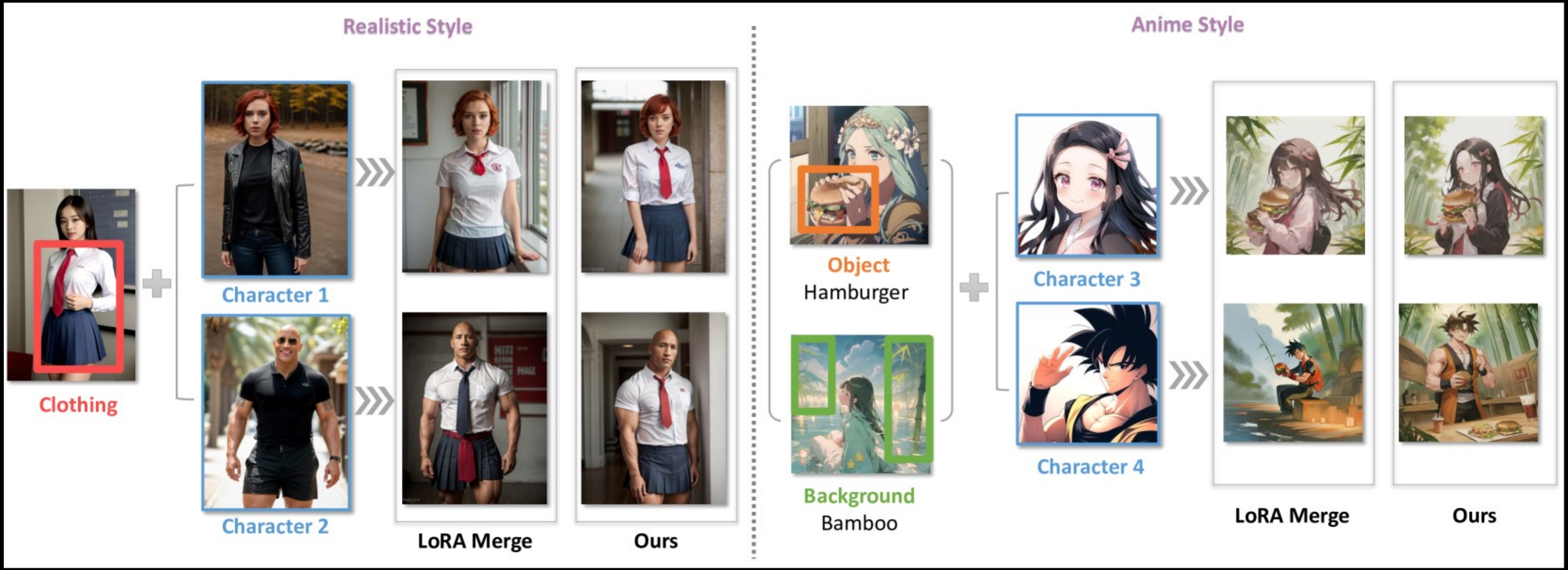
drkfnts style

Trigger words:

Anime,
Cyberpunk

COMBINING MULTIPLE LORAS

- As LORAs are "nudges" to the model output, we can combine multiple LoRAs and combine the effects!



Multi-LoRA Composition for Image Generation, Zhong et.al, 2024

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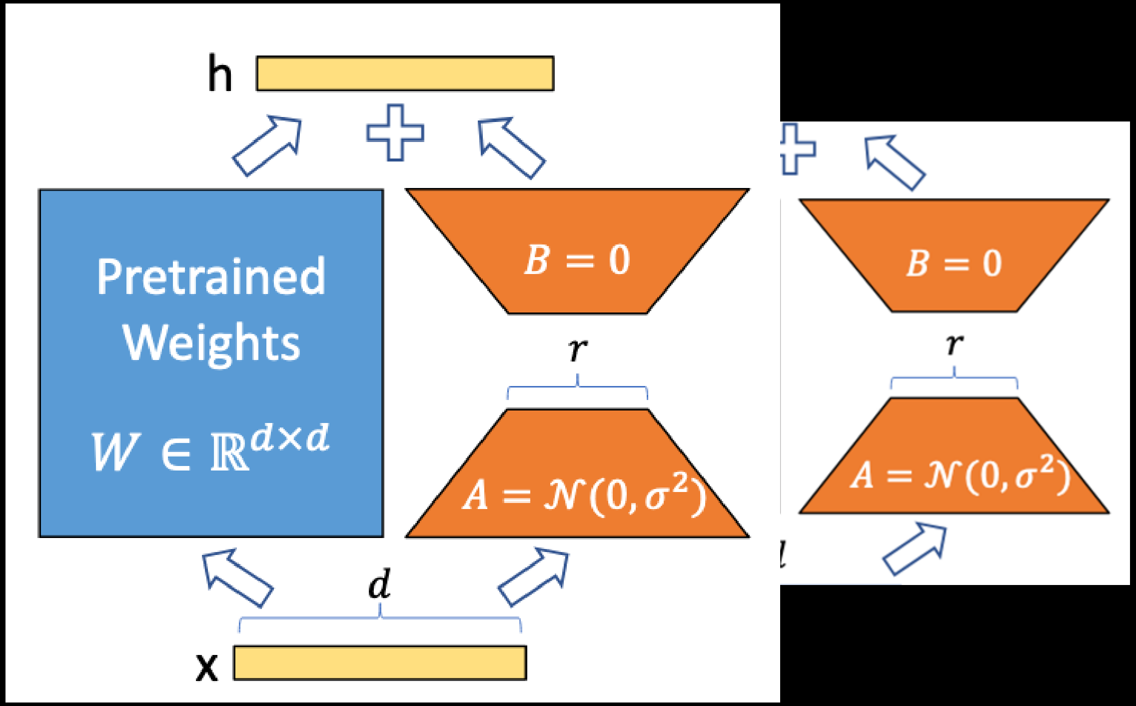


Multi-LoRA Composition for Image Generation, Zhong et.al, 2024

MODEL SPECIFIC

- Since LoRAs are all techniques to fine-tune an existing model, they are model specific.
- Methods trained on a specific model will not work with a different model, so it's important to keep the exact settings.
- This could include VAE, text encoder, or even the quantization level of the model!

FINE-TUNED MODELS

- Fine-tuning the model often gives the best results than prompting with images.
- This requires a base model trained on a giant set of training data, in which anyone can customize the model for their specific purposes.
- On Oct 2022, Runway ML released the weights for SD1.5, that started the explosion of high-quality image generation models and experimentation.

BASE MODELS

- Stable Diffusion 1.5 (Oct 2022)
- Stable Diffusion 2.0 (Nov 2022)
- Stable Diffusion XL (Jul 2023)
- Stable Diffusion 3 (Feb 2024)
- FLUX (Aug 2024)
- FLUX 2.0 (Nov 2025)
- Z-Image (Dec 2025)

MODEL VARIANTS AND MERGING

- Since the base models were accessible to the public, we have many variants, or "checkpoints" that fine-tune the model for specific stylistic effects or purpose.
- People found out you could also "merge" different checkpoints from the same base models to combine the strengths of different models.
- As these models are privately trained, they are trained with data with no restrictions. Some of which could raise ethical, legal concerns.

Juggernaut XL:
Photogenic Images

Pony Diffusion:
Anthro Characters

Animagine XL:
Anime Aesthetics

ETHICS OF RELEASING BASE MODELS

- These release of public weights is a double-edged sword.
- On one hand it allowed an explosion of techniques and discoveries on the image generation by providing access to models that are too expensive to train from scratch for any individuals.
- But it also allowed explosions of image generation models that are unethical in its training data collection, specific subject matter, or commercial usage when the public consensus and legal systems are not yet ready.

- This debate is not limited to image generation. It will be echoed throughout various domains.

RETRIEVAL-AUGMENTED GENERATION
VECTOR DATABASE

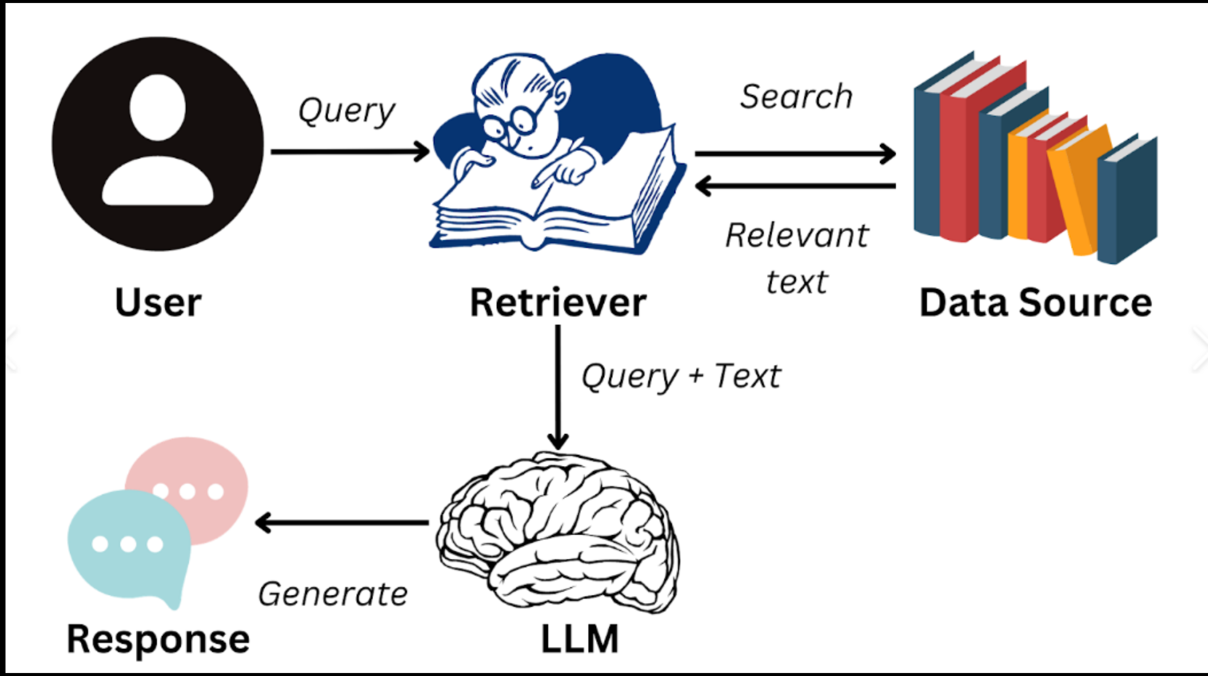
ISDN 3150 AI for Design, Yuan, 2026 Spring

INTRODUCTION

- Overview of Retrieval-Augmented Generation (RAG)
- Overview of Vector Database
  - Embeddings
  - Approximate Nearest Neighbors
- MongoDB

RETRIEVAL-AUGMENTED GENERATION

- Retrieval-Augmented Generation (RAG) is the process of optimizing the output of a large language model, so it references an authoritative knowledge base outside of its training data sources before generating a response.



EXAMPLES OF RAG

- The Ultimate "Rules Lawyer" for Tabletop Games
- A chatbot that settles arguments during complex games like *Dungeons & Dragons* or *Magic: The Gathering*.
- **The Problem:** Standard AI hallucinates game rules or mixes up different editions of the game.
- We upload the official 300-page rulebook PDF into the system's database. When a player asks, "Does casting a fireball underwater do half damage?", the system first **retrieves** the specific paragraphs about underwater combat and fire magic. It then **generates** an answer grounded strictly in those paragraphs, citing the exact page number.

EXAMPLES OF RAG

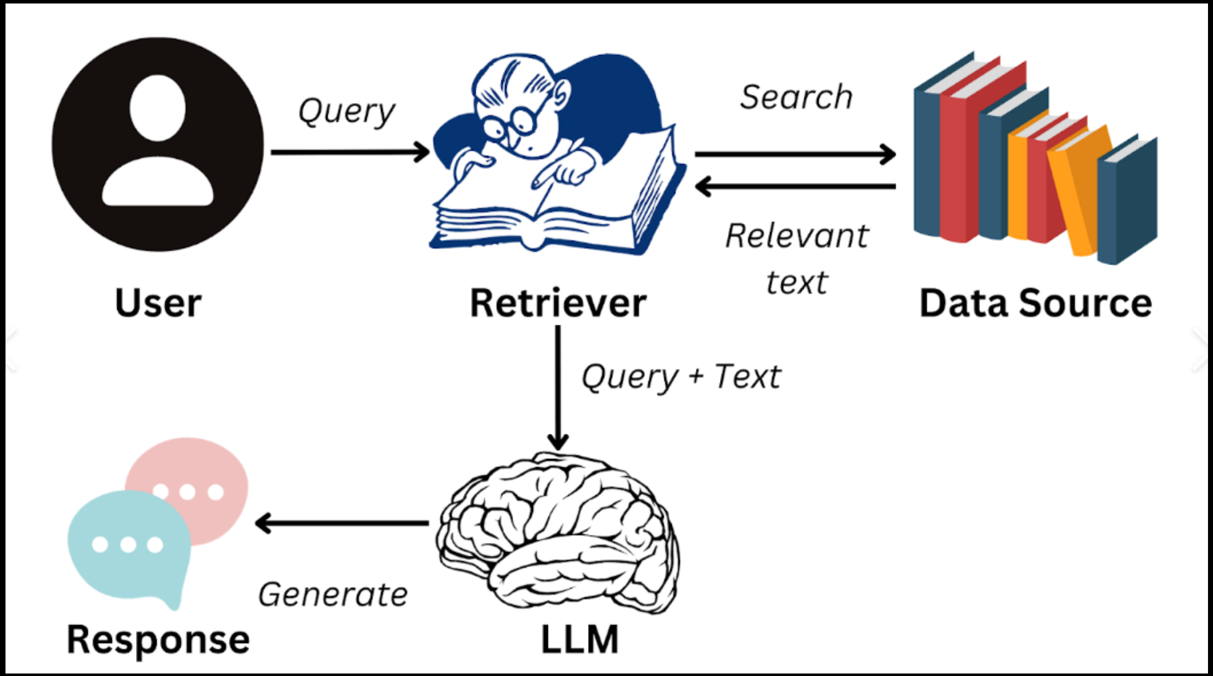
- The "What’s in my Fridge?" Chef
- A recipe generator that actually understands the struggle of a student's limited pantry and doesn't suggest buying expensive ingredients.
- **The Problem:** If you ask a standard AI for a recipe, it will give you a generic one that requires a trip to the grocery store.
- The database is filled with thousands of quick, cheap recipes. The user snaps a photo of their open fridge or types in three ingredients (e.g., "eggs, rice, hot sauce"). The system **retrieves** only the recipes from the database that primarily use those exact ingredients. It then **generates** a customized, step-by-step cooking guide combining those retrieved recipes with a fun, encouraging tone.

EXAMPLES OF RAG

- The Video Game NPC with a "Memory“
- Giving a Non-Playable Character (NPC) in a video game a unique backstory and a continuous memory of everything the player has ever done. **How it works:**
- **The Problem:** Standard NPCs just repeat the same three lines of dialogue.
- The database holds two things: the NPC’s "lore bible" (their personality, secrets, and goals) and a running transcript of every interaction the player has had with them. When the player walks up and says "Hello," the system **retrieves** the NPC's personality *and* the memory that the player stole an apple from them yesterday. The LLM then **generates** a dynamic response: *"Oh, it's you. I hope you're not here to steal more of my fruit."*

BENEFITS OF RAG

- Cost-effective implementation
- Current information
- Enhanced user trust
- More developer control



WORKFLOW OF RAG

- Create External Data:
    - External data is sourced from APIs, databases, or document repositories.
    - Data is converted into numerical representations using embedding language models, stored in a vector database.

- Retrieve Relevant Information:
    - User queries are converted to vectors and matched against the vector database.
    - Relevant documents are retrieved based on mathematical vector calculations.

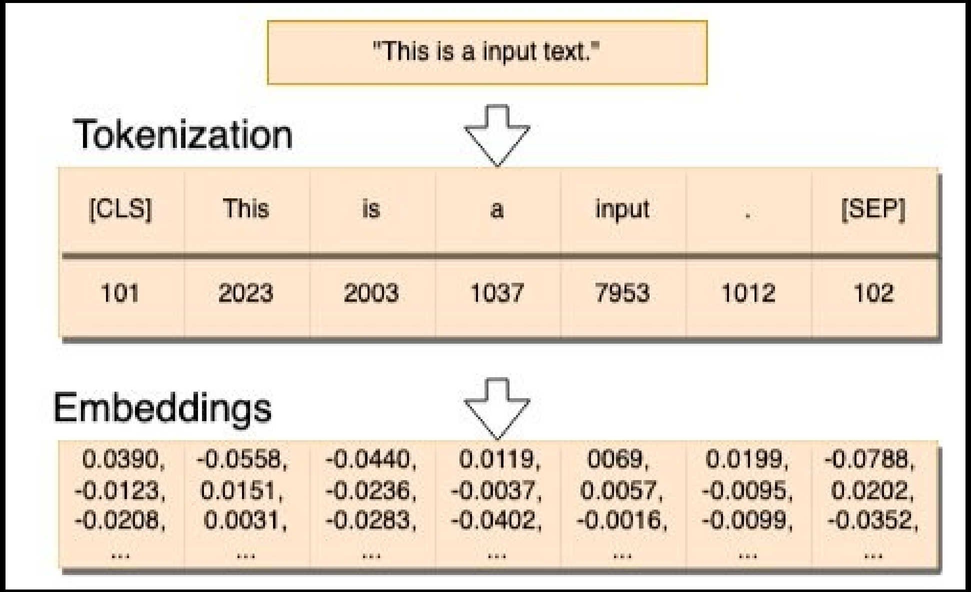
- Augment the LLM Prompt:
    - The RAG model adds retrieved data to the user input for better context.
    - Uses prompt engineering to improve LLM responses.

- Update External Data:
    - Maintain current data by updating documents and their embeddings.
    - Use automated or periodic processes to handle data updates.

WHAT IS VECTOR DATABASE?

- A vector database is a specialized type of database optimized for storing and retrieving high-dimensional vector embeddings.

- An embedding is a numerical representation of a piece of information.



EXAMPLE

- An e-commerce website with 1 million products wants to provide personalized recommendations.
- How to find related products based on user queries?

Exact Match (Keywords Match) doesn't work :

BENEFIT OF VECTOR DATABASE

- Traditional databases and search engines are not designed to handle the complexity and high dimensionality of modern data, such as images, videos, and text embeddings.
- Vector databases index data as vectors and use similarity search algorithms to efficiently search for and retrieve similar data points based on their distance or similarity to a query point.
- This enables faster and more accurate analysis of high-dimensional data, unlocking the potential for a wide range of applications, such as e-commerce recommendations, autonomous vehicle systems, and natural language processing.
- Vector databases can also be easily scaled to handle large amounts of data and traffic and can be integrated with a variety of existing systems and platforms.

DATA TYPES AND SOURCES

- Data Types and Sources

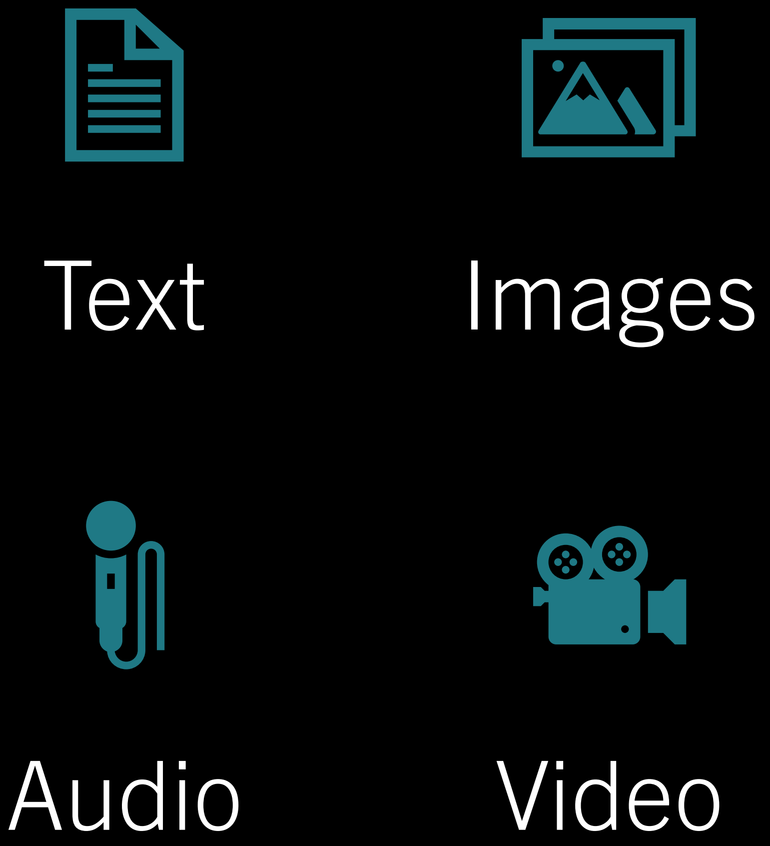
  Text, images, audio, video

- Techniques for Generating Embeddings

  Word embeddings

  Image embeddings

  Custom embeddings using deep learning models



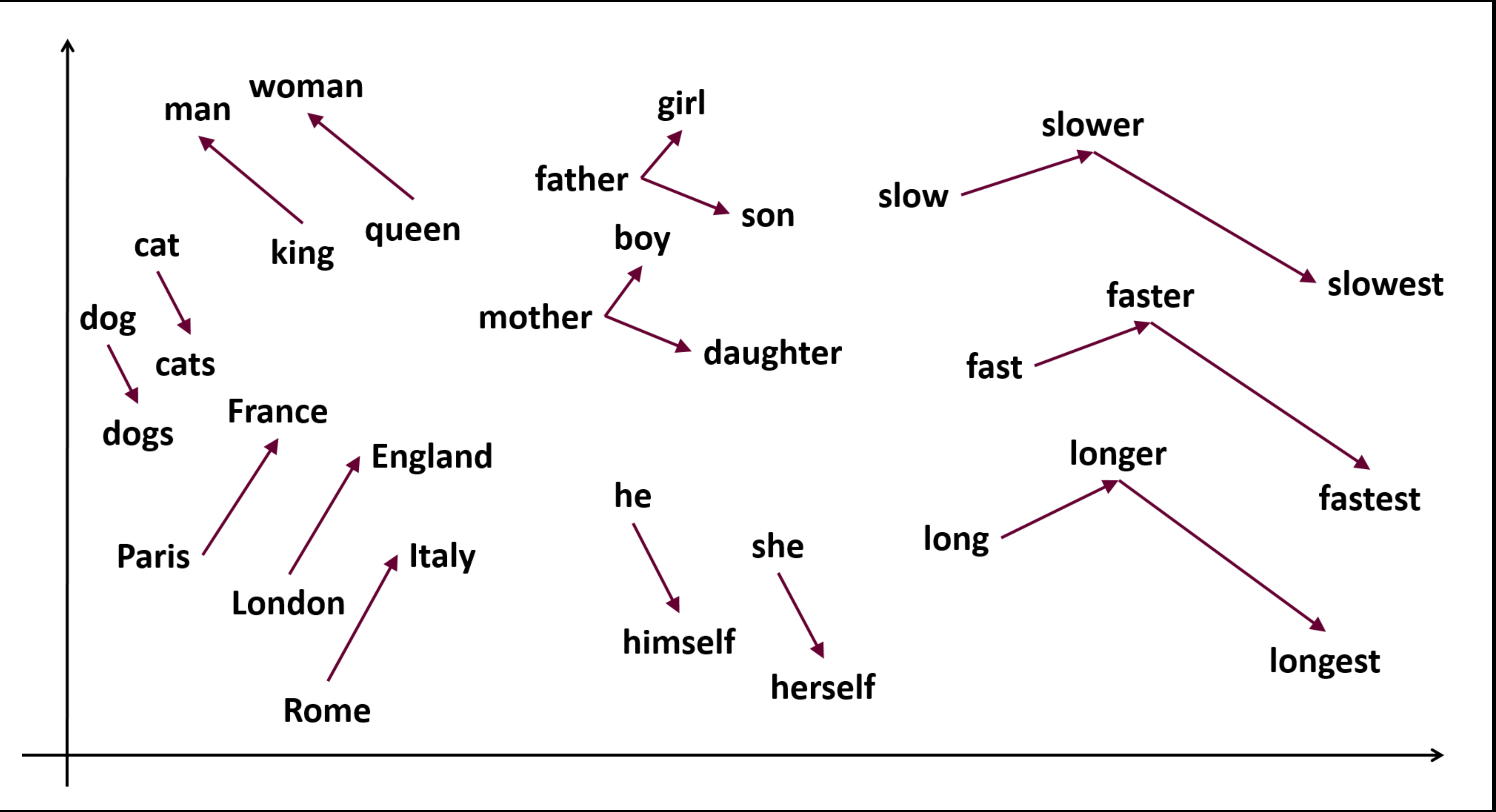
WORD EMBEDDINGS

- Word embeddings are a way of representing words as vectors in a multi-dimensional space, where the distance and direction between vectors reflect the similarity and relationships among the corresponding words.

Frequency-based embeddings:
Term Frequency-Inverse Document Frequency (TF-IDF), TF-IDF is designed to highlight words that are both frequent within a specific document and relatively rare across the entire corpus, thus helping to identify terms that are significant for a particular document.

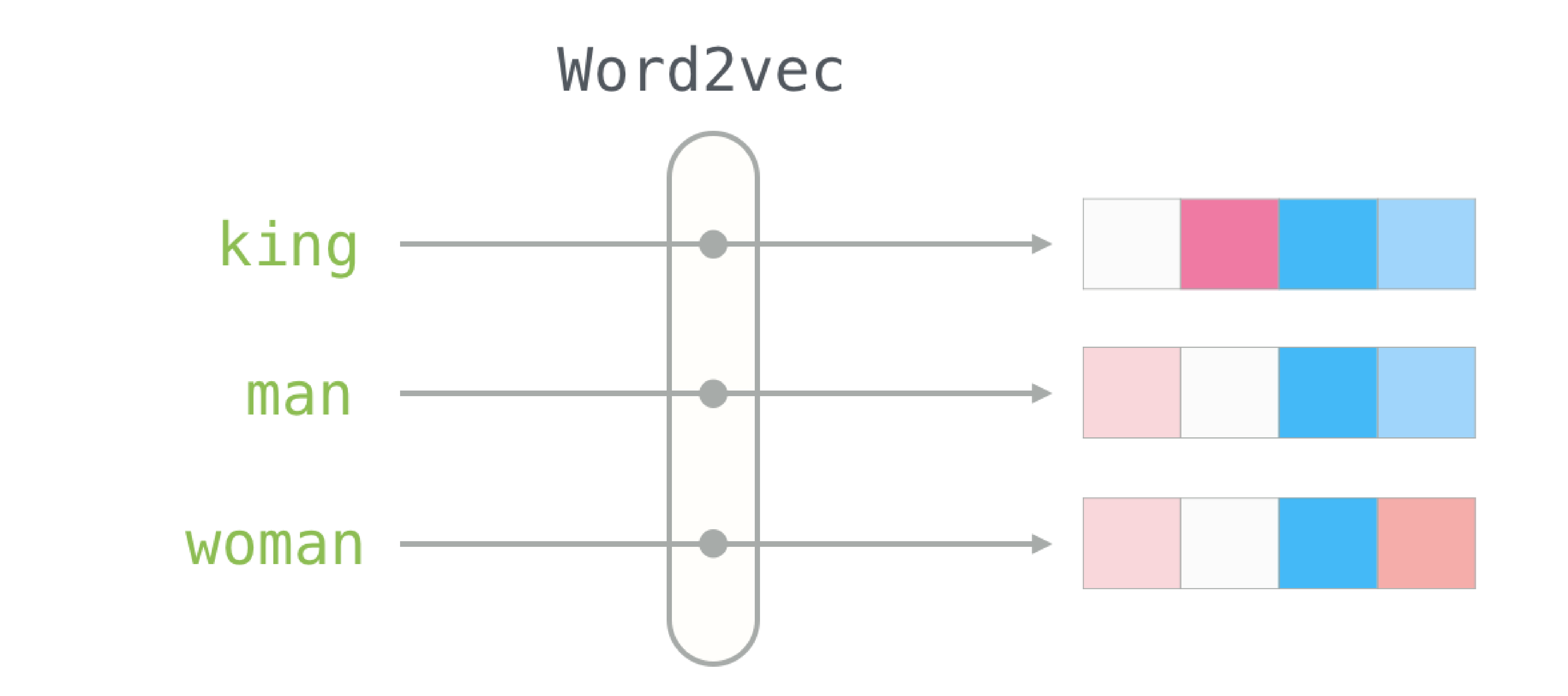
Prediction-based embeddings
Prediction-based embeddings are word representations derived from models that are trained to predict certain aspects of a word's context or neighboring words.

WHAT WE EXPECT



CAN WE DO THIS MORE EFFICIENT?

- Word2Vec



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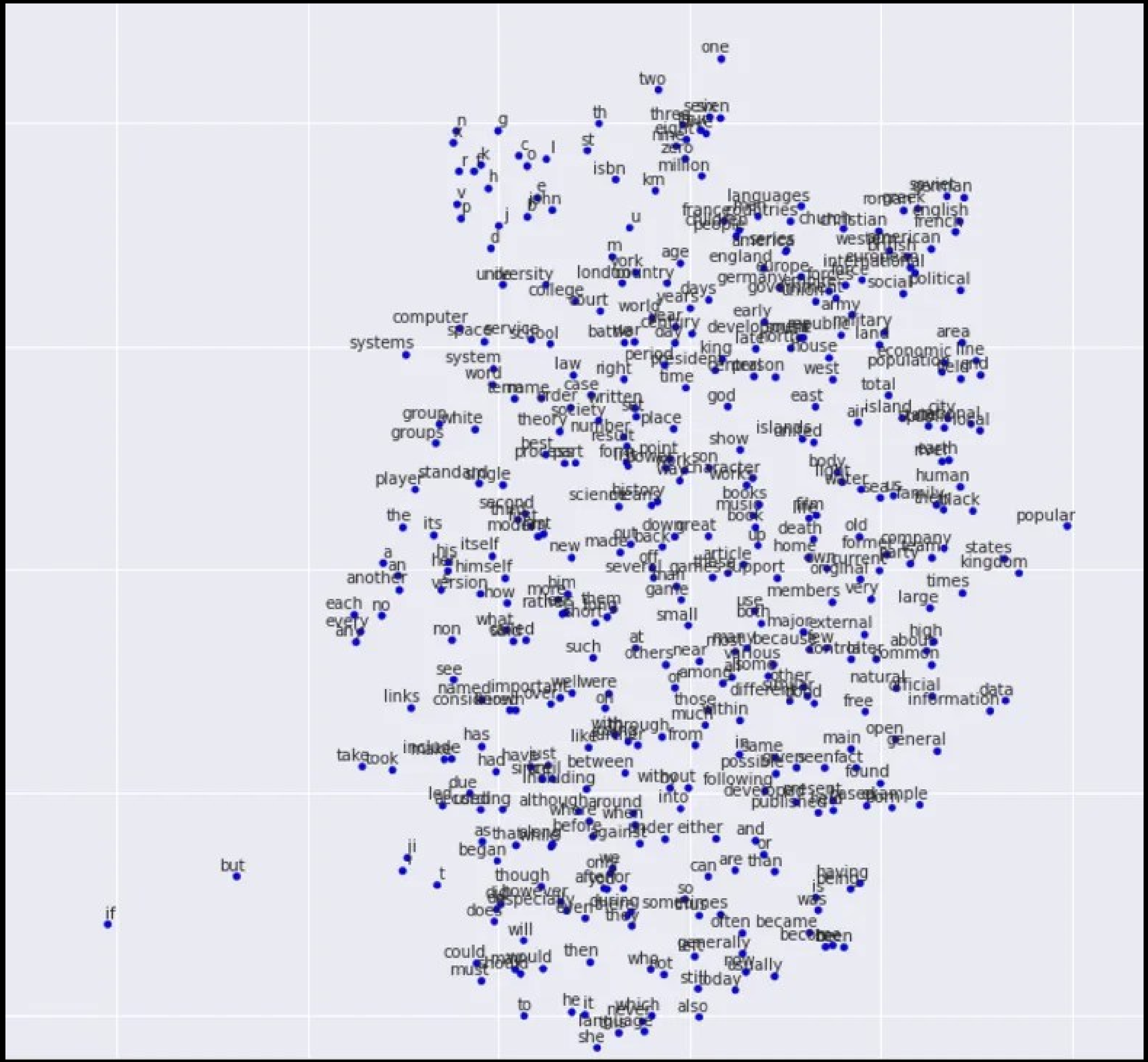
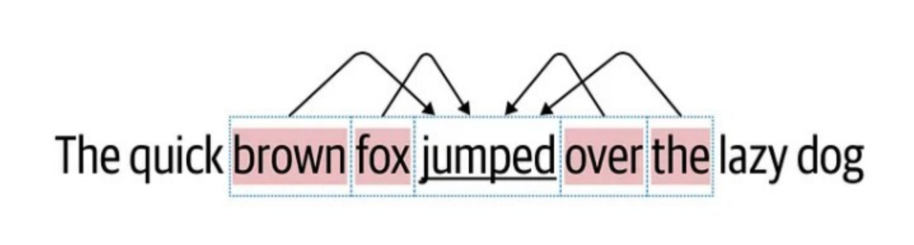


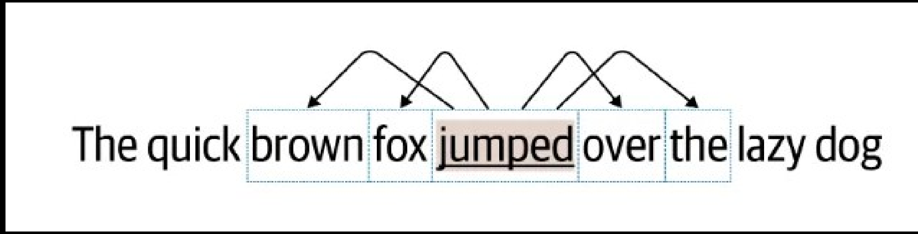
WORD2VEC

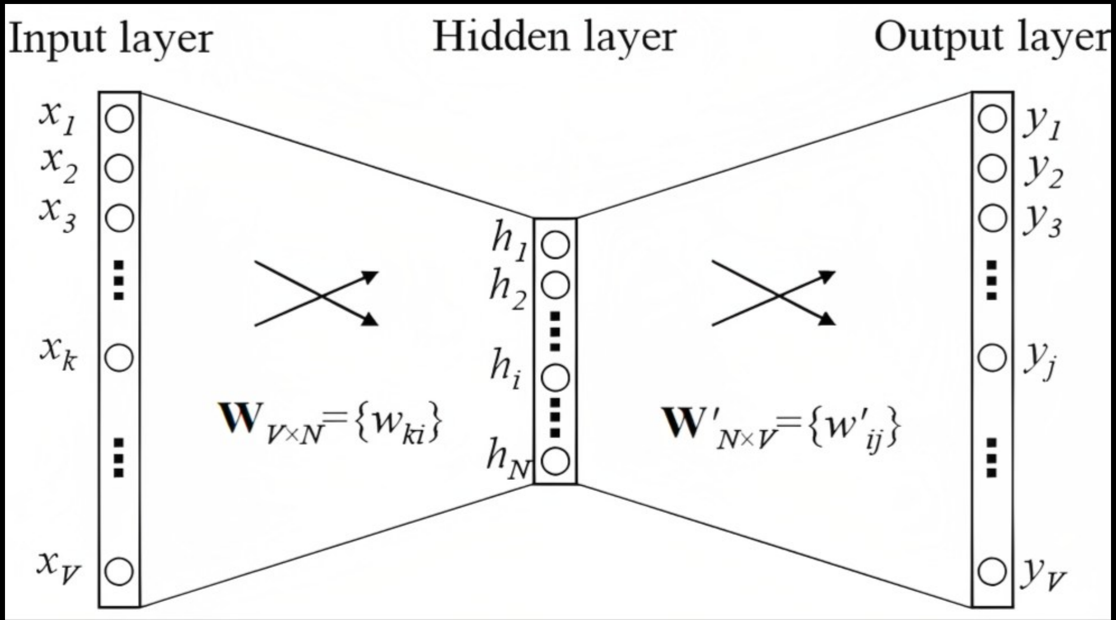
- Word2Vec consists of two main models for generating vector representations: Continuous Bag of Words (CBOW) and Continuous Skip-gram.

Continuous Bag of Words (CBOW) model aims to predict a target word based on its surrounding context words within a given window. It uses the context words to predict the target word, and the learned embeddings capture semantic relationships between words.



Continuous Skip-gram model takes a target word as input and aims to predict the surrounding





GLOBAL VECTORS FOR WORD REPRESENTATION

- GloVe is a word embedding model designed to capture global statistical information about word co-occurrence patterns in a corpus.
- GloVe is based on the idea that the global statistics of word co-occurrence across the entire corpus are crucial for capturing word semantics. It considers how frequently words co-occur with each other in the entire dataset rather than just in the local context of individual words.

GLOBAL VECTORS FOR WORD REPRESENTATION

- Let the matrix of word-word co-occurrence counts be denoted by X, whose entries X_{ij} tabulate the number of times word j occurs in the context of word i.
- Let $X_i = \sum_k X_{ik}$ be the number of times any word appears in the context of word i.
- Finally, let $P_{ij} = \mathrm{P}[j|i] = \frac{X_{ij}}{X_i}$ be the probability that word j appear in the context of word i.

Probability and Ratio	$k = solid$	$k = gas$	$k = water$	$k = fashion$
$P(k\|ice)$	1.9×10^{-4}	6.6×10^{-5}	3.0×10^{-3}	1.7×10^{-5}
$P(k\|steam)$	2.2×10^{-5}	7.8×10^{-4}	2.2×10^{-3}	1.8×10^{-5}
$P(k\|ice)/P(k\|steam)$	8.9	8.5×10^{-2}	1.36	0.96

Co-occurrence probabilities for target words ice and steam with selected context words from a 6 billion token corpus

BEYOND WORD2VEC AND GLOVE

Subword embeddings, such as FastText, represent words as combinations of subword units, providing more flexibility and handling rare or out-of-vocabulary words.

For example, the word "unhappiness" can be decomposed into "un," "happy," and "ness."

Attention **mechanisms and transformer models** consider contextual information and bidirectional relationships between words, leading to more advanced language representations. Attention mechanisms were introduced to improve the ability of neural networks to focus on specific parts of the input sequence when making predictions.

TECHNIQUES FOR GENERATING EMBEDDINGS

- Word Embeddings
- Sentence Embeddings
- Document Embeddings
- Image Embeddings

Convolutional Neural Networks (CNN), Transformer.

VGG-16, ResNet50, InceptionV3, NobileNetV2

SEARCH FOR SIMILAR VECTORS

- Similarity Metrics

  Cosine similarity

  Euclidean distance

  Jaccard index

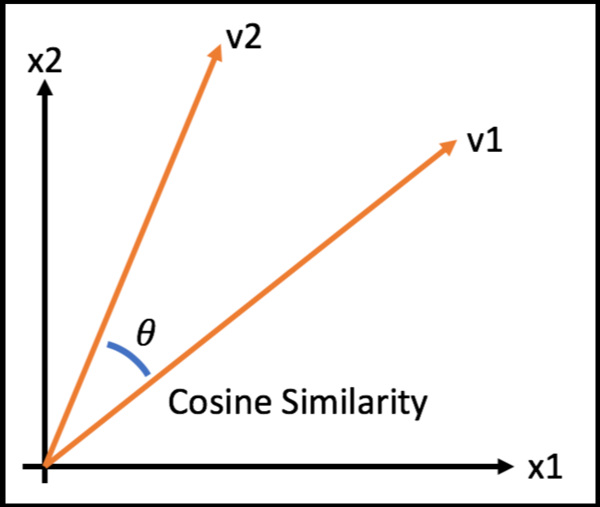
- Querying Techniques

  Exact match vs. similarity search

SIMILARITY METRICS

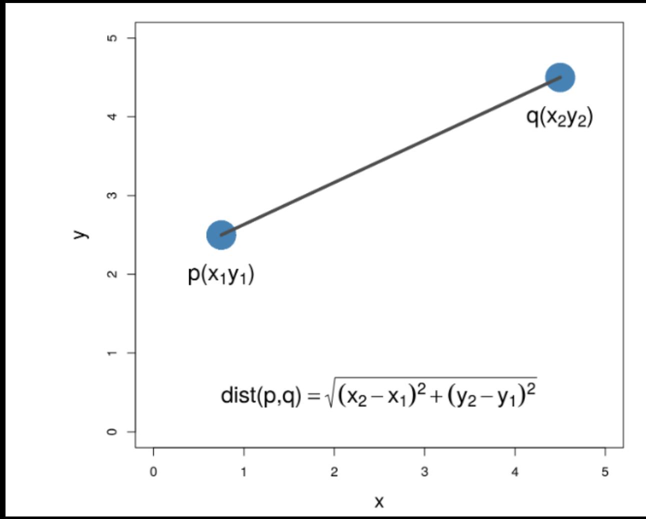
- Cosine similarity

- $S_C(A,B) = \cos(\theta) = \frac{A \cdot B}{\|A\|\|B\|} = \frac{\sum_{i=1}^{n} A_i B_i}{\sqrt{\sum_{i=1}^{n} A_i^2} \cdot \sqrt{\sum_{i=1}^{n} B_i^2}}$



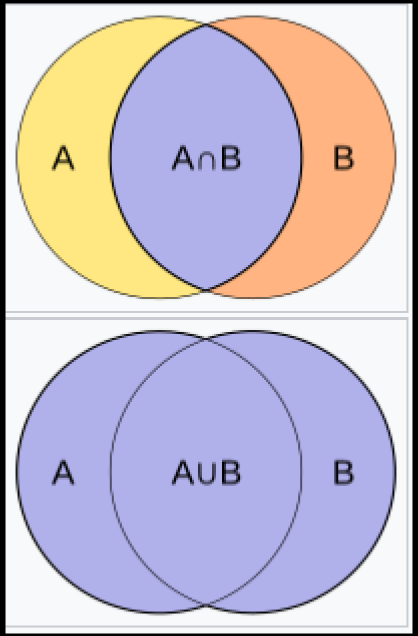
- Euclidean distance

- $d(p,q) = \|p - q\|$

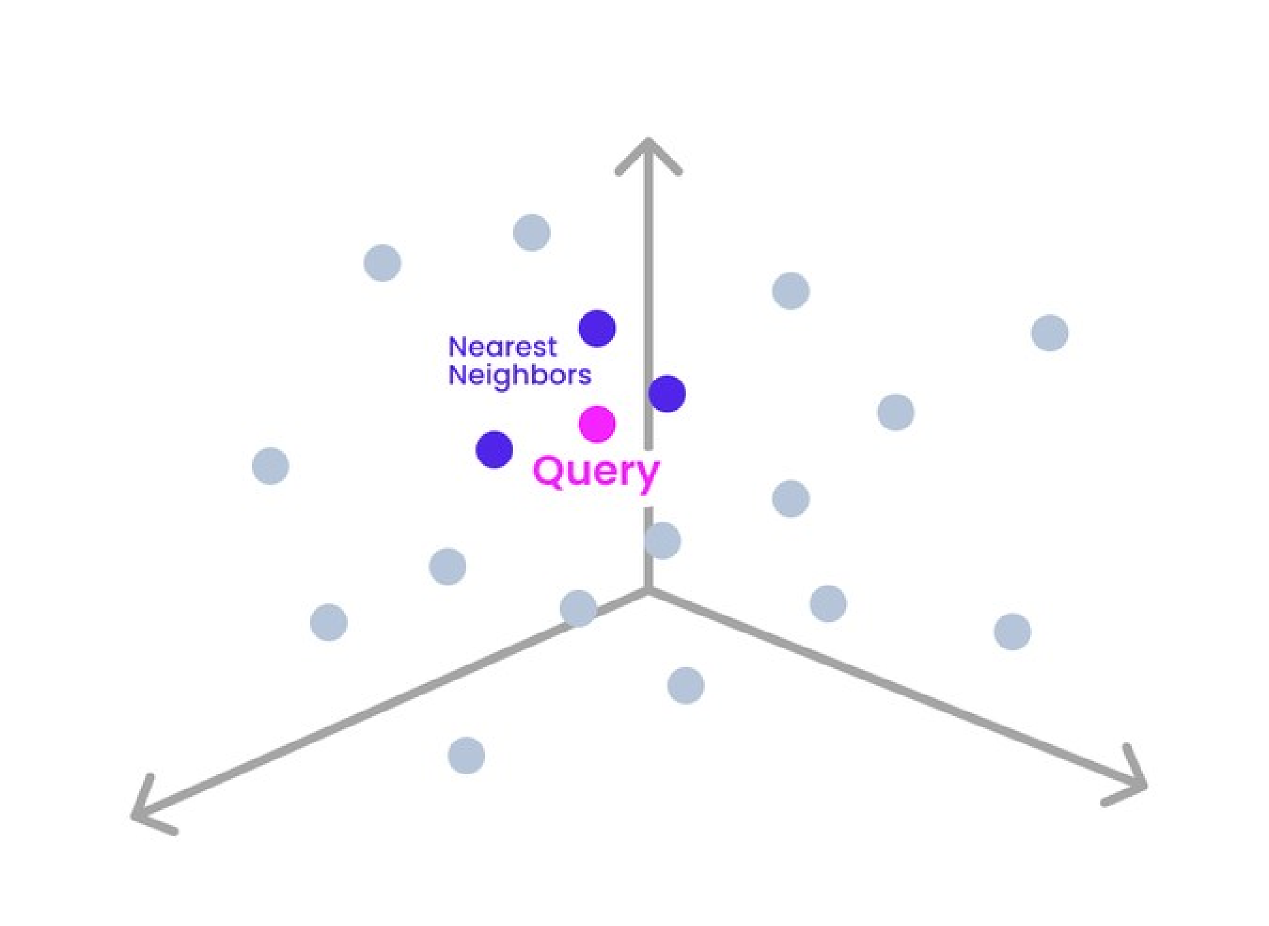


- Jaccard index

$$J(A,B) = \frac{|A \cap B|}{|A \cup B|} = \frac{|A \cap B|}{|A| + |B| - |A \cap B|}.$$



VECTOR SEARCH



EXACT K-NEAREST NEIGHBORS ALGORITHM

- k-Nearest Neighbors is a simple algorithm that finds the k exact nearest neighbors of a given query point (observation).
- Steps:
  1. Choosing k
  2. Choosing a distance measurement
  3. The actual query run on your vector dataset
  4. The decision made to accomplish your specific task, based on the results of your query

O(n) time complexity, where n denotes the total number of data points.

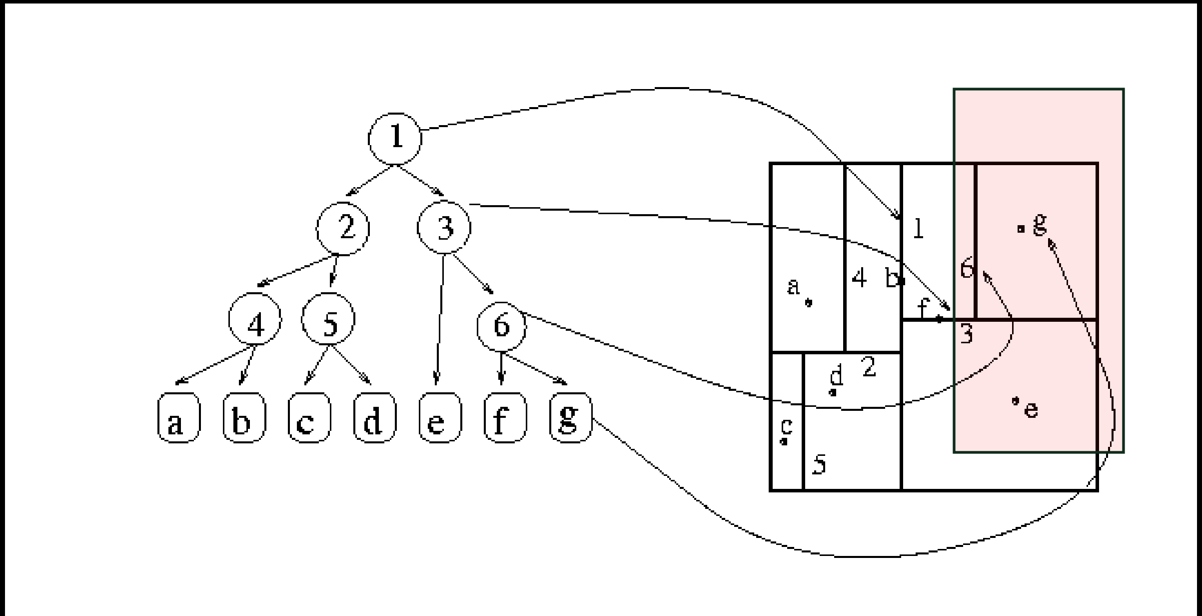
APPROXIMATE NEAREST NEIGHBORS

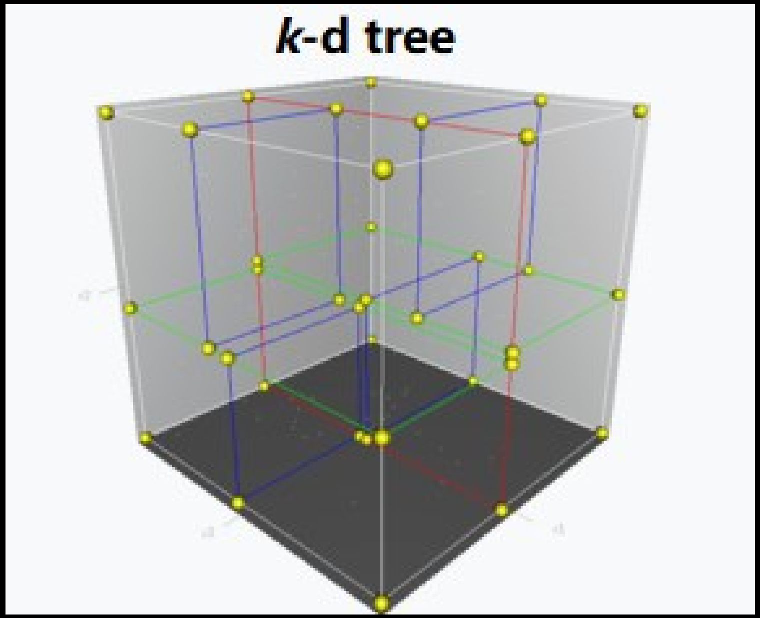
- In situations where exact matches aren't critical, Approximate Nearest Neighbor (ANN) search offers a performance-optimized alternative to traditional k-NN search. Unlike traditional methods that require examining the entire dataset to find the nearest neighbor, ANN algorithms can skip parts of the search space or stop once a satisfactory match is found. This results in faster searches and reduced computational demands.

- K-dimensional tree
- Locality-sensitive hashing (LSH)
- Product quantization
- Hierarchical Navigable Small Worlds (HNSW)

K-DIMENSIONAL TREE

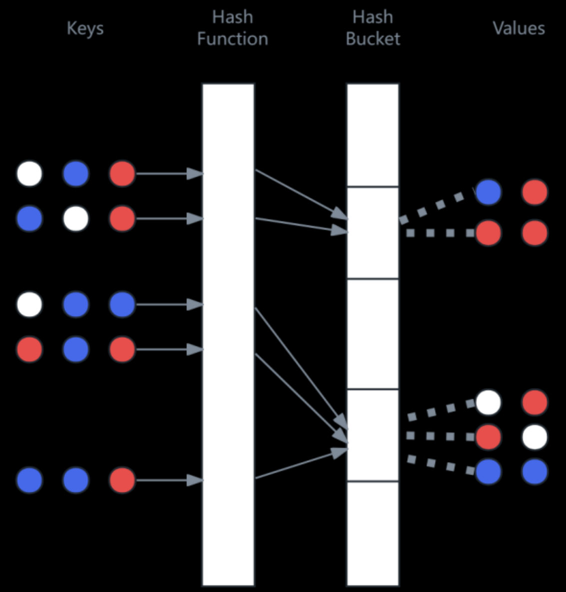
k-dimensional tree is a data structure used for organizing points in a k-dimensional space.





LOCALITY-SENSITIVE HASHING (LSH)

Locality Sensitive Hashing (LSH) is a technique that efficiently approximates similarity search by reducing the dimensionality of data while preserving local spatial relationship between points.
The core idea behind LSH is to hash data points in such a way that similar items are mapped to the same hash buckets with high probability.



Step 1: Hashing

- Create multiple hash functions to generate binary hash codes for each data point.

Step 2: Binning

- Distribute data points into hash buckets based on their hash codes, grouping similar points together.

Step 3: Querying

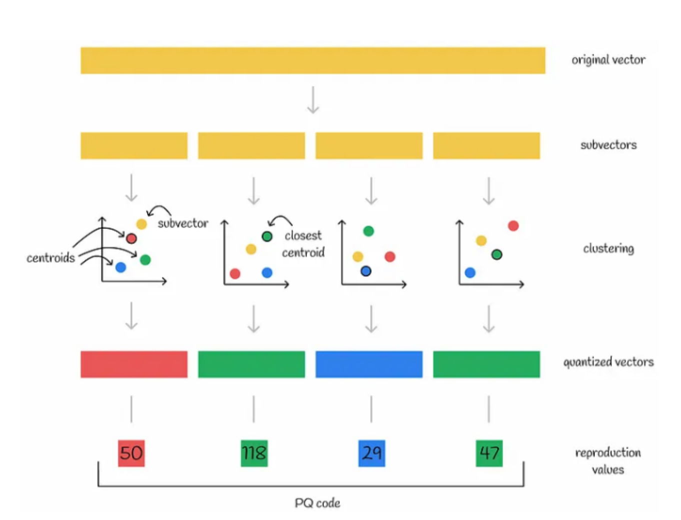
PRODUCT QUANTIZATION

1. Taking a big, high-dimensional vector

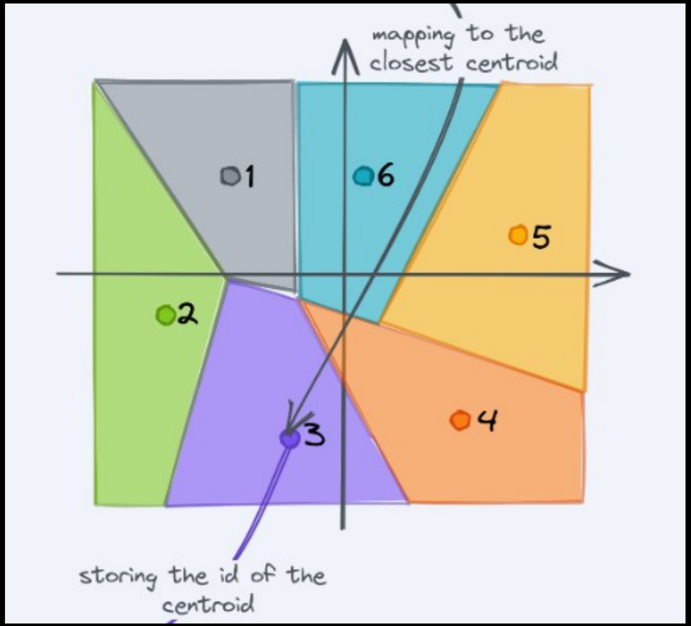
2. Splitting it into equally sized chunks — our subvectors

3. Assigning each of these subvectors to its *nearest centroid* (also called reproduction/reconstruction values),

4. Replacing these centroid values with unique IDs — each ID represents a centroid.



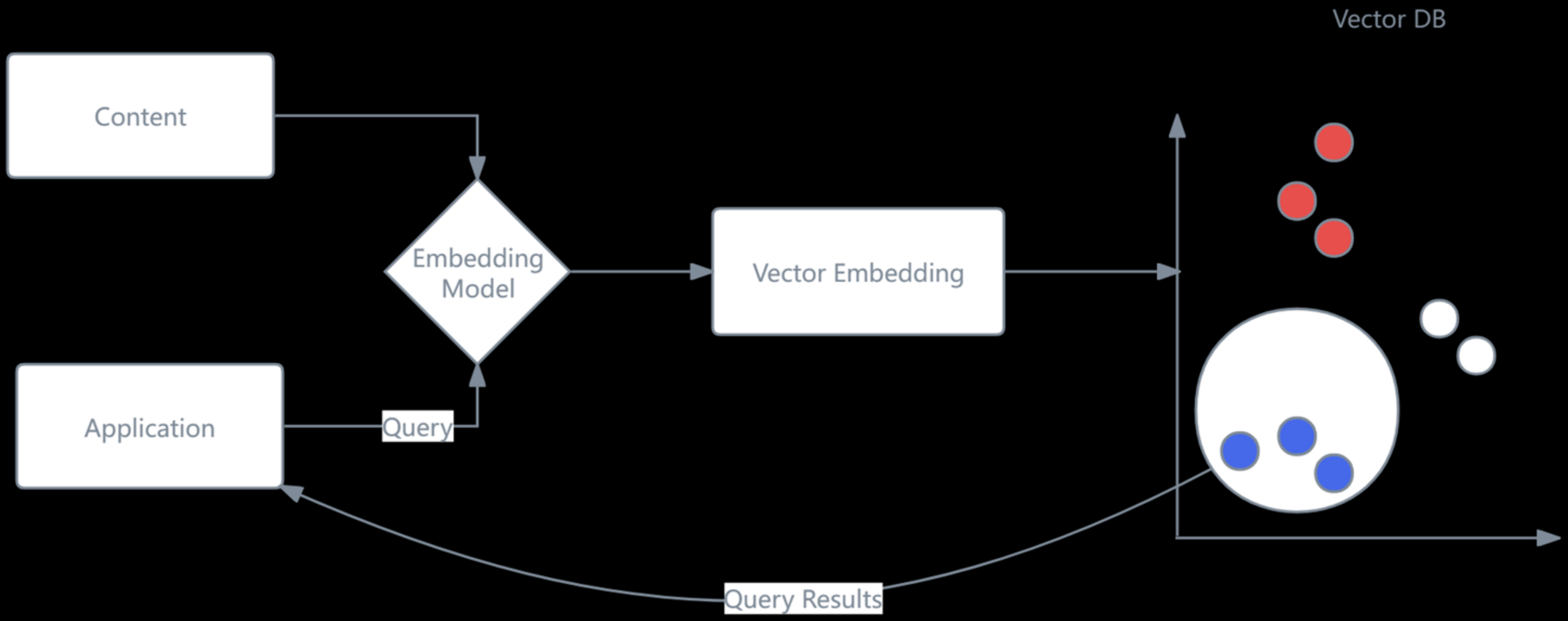
Encoding using quantization



VECTOR DATABASE

ISDN 3150 AI for Design, Yuan, 2026 Spring

ROLE OF VECTOR DATABASES

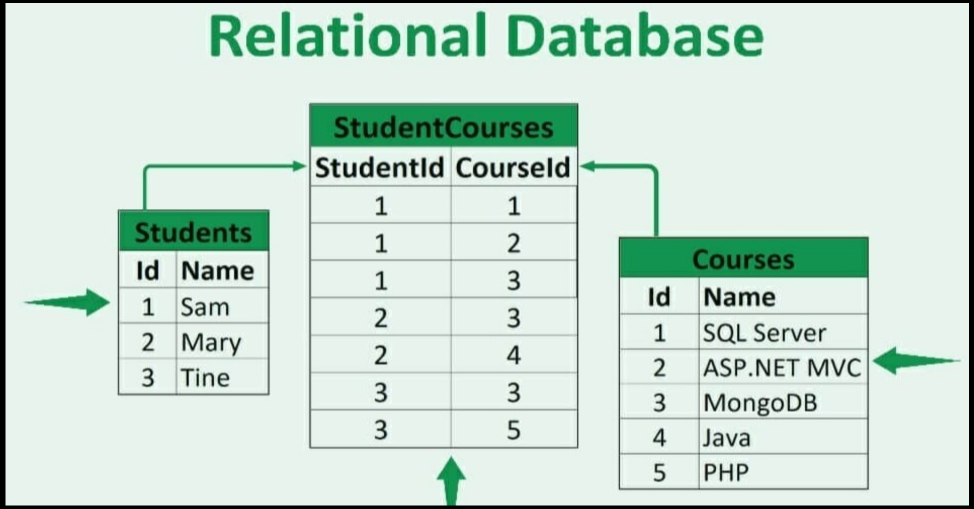


MONGO DB

MongoDB is a popular open-source NoSQL database designed for modern applications that require scalability, flexibility, and high performance.

Unlike traditional relational databases, MongoDB stores data in a document-oriented format, using BSON (Binary JSON) to represent data structures.


```
_id: ObjectId('670f6a47c5142279b8e965b6')
listing_url : "https://www.airbnb.com/rooms/10021707"
name : "Private Room in Bushwick"
summary : "Here exists a very cozy room for rent in a shared 4-bedroom apartment.…"
space : ""
description : "Here exists a very cozy room for rent in a shared 4-bedroom apartment.…"
neighborhood_overview : ""
notes : ""
transit : ""
access : ""
interaction : ""
house_rules : ""
property_type : "Apartment"
room_type : "Private room"
bed_type : "Real Bed"
minimum_nights : 14
maximum_nights : 1125
```

RAG SUMMARY

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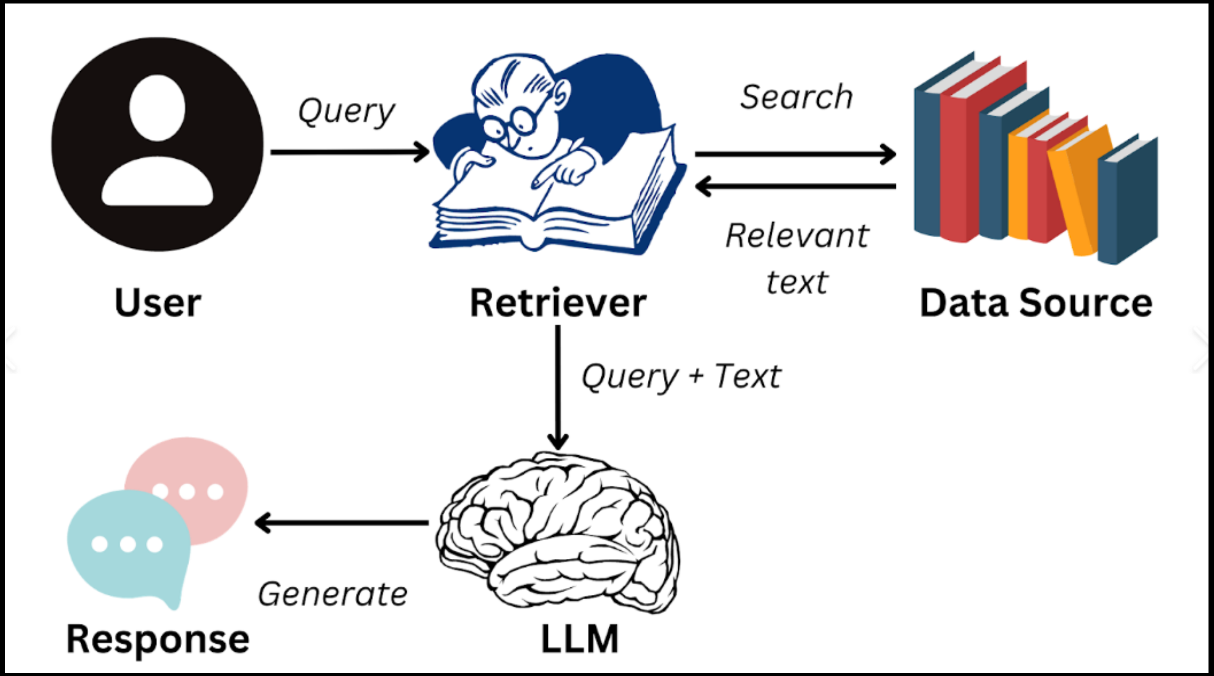


FINAL PROJECT

ISDN 3150, Yuan, 2026 Spring

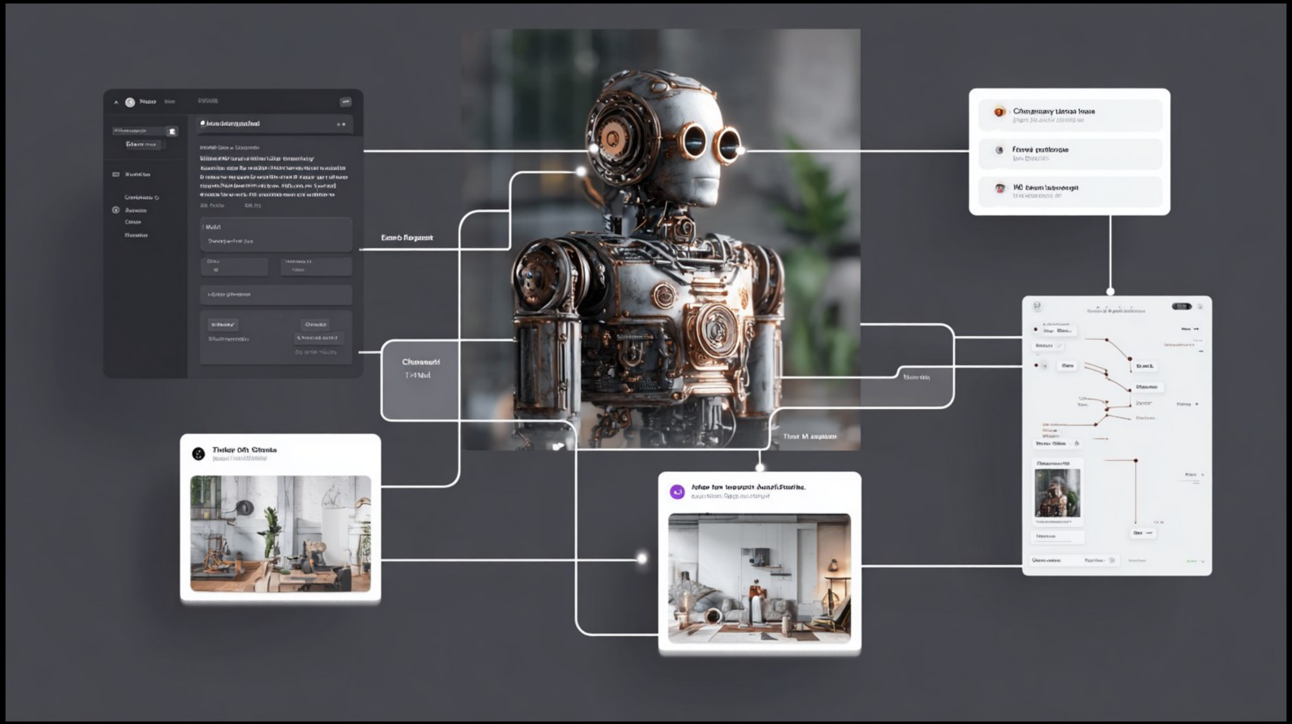
AI-POWERED INTERACTIVE DESIGN PROTOTYPE

Project Objective & Requirements

- You will design and develop **a functional, interactive Web App or Website** (using frameworks like Streamlit) powered by the AI tools we have learned this semester.
- Your prototype should not just be a static presentation; it must be a "playable," interactive system that solves a specific problem, empowers a design workflow, or offers a highly creative and engaging user experience.
- I don't put any topic constraints here! Be creative!

AI-POWERED INTERACTIVE DESIGN PROTOTYPE

- Examples
- **The "AI Client" Co-Creator**: Build an Agent with a highly specific persona (e.g., a time-traveling Victorian inventor). Users chat with this agent to uncover their "needs." Once the agent is satisfied, it uses Function Calling to trigger an image generation API, creating a customized product render directly in the chat interface. – **Puzzle Solving Game?**



AI-POWERED INTERACTIVE DESIGN PROTOTYPE

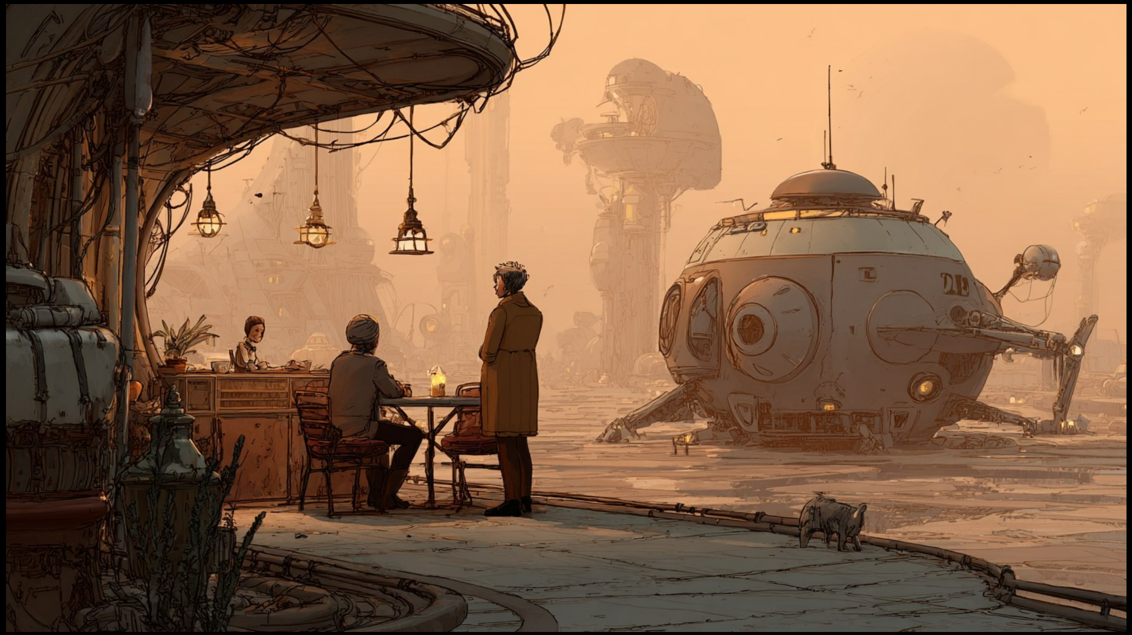
- Examples
- **The "Gacha Toy" Universe Generator**: A blind-box style web app. Users input random, unrelated keywords (e.g., "Astronaut + Sushi + Cyberpunk"), and the system uses RAG (drawing from a database of vinyl toy materials and styles) combined with a customized Stable Diffusion model to generate a cohesive toy IP, complete with a backstory.

AI-POWERED INTERACTIVE DESIGN PROTOTYPE

- Examples
- **Cross-Domain Mashup Designer:** A tool that forces creative cross-pollination. The user selects two completely unrelated fields (e.g., Deep Sea Biology and Kitchen Appliances). The system utilizes multi-agent RAG to extract features from both domains and generates a novel, hybrid product concept.

AI-POWERED INTERACTIVE DESIGN PROTOTYPE

- Examples
- The Interactive World-Builder & NPC Ecosystem (Game/Community Design): A web app acting as a dynamic sandbox for game designers or virtual community managers. The user inputs a foundational theme (e.g., "A collaborative solarpunk settlement on Mars"). The system utilizes Agents to simulate different community members or factions, each with distinct personas and needs. – Market Simulator?



AI-POWERED INTERACTIVE DESIGN PROTOTYPE

- Milestone 1: Proposal Presentation (15 minutes per team) (Apr. 20th)
- Pitch your core concept: What is the problem or experience you are designing?
- Target audience and expected user journey.
- Proposed technical architecture.

AI-POWERED INTERACTIVE DESIGN PROTOTYPE

- Milestone 2: Final Presentation & Live Demo (30 minutes per team) (May 11th)
- A compelling pitch of the final product.
- Live Trial Session: This is mandatory. You will provide a link or device for the class and guest critics to physically interact with your Web App/Website. Real-time generation and interaction will be a key highlight.

AI-POWERED INTERACTIVE DESIGN PROTOTYPE

- Final Submission:
- Project Source Code: All Python scripts, asset folders, and a requirements.txt file. The code should be well-commented.
- Presentation Slides: The slides used for your final pitch.
- Technical Documentation (System Architecture Report): A concise document (2-3 pages) explaining how you utilized the course tools. You must detail your prompt engineering strategies, how RAG/Function Calling was implemented, which image models were used, and how the technical pipeline aligns with your design goal.

AI-POWERED INTERACTIVE DESIGN PROTOTYPE

- Grading Rubric (30 Points Total)
- Proposal Presentation (5 Points): Clarity of the initial concept, creativity of the problem space, and feasibility of the proposed technical approach.
- Final Presentation & Live Demo (5 Points): Quality of storytelling, UX/UI of the prototype interface (Streamlit/Gradio), and the smoothness of the live interactive trial.
- Technical Depth & Execution (15 Points): How comprehensively and effectively did you integrate the taught stack? Does the code work as intended? Is the output (text and image) of high quality and relevant to the user's input?
- External Expert Evaluation (5 Points): We will invite external experts to the final review of your final presentations. They will score your project based on its overall innovation, design value, and user engagement during the live demo.

THANKS FOR ATTENDING LECTURE!

AI for Design

Yuan Liu